

Scenario 1: EV Battery Fault Detection

An AI model predicts whether an EV battery has a **fault**.

Out of 100 batteries:

- 30 actually have faults.
- The model correctly detects 25 faulty batteries.
- It incorrectly identifies 10 healthy batteries as faulty.

Questions:

Q1. What is the True Positive (TP)?

Q2. What is the False Positive (FP)?

Q3. What is the False Negative (FN)?

Q4. What is the True Negative (TN)?

Confusion Matrix

	Predicted Fault	Predicted Healthy
Actual Fault	TP = 25	FN = 5
Actual Healthy	FP = 10	TN = 60

Ans1

Answer: 25

2

Answer: 10

3

Actual faulty batteries = 30

Detected faulty = 25

FN = 30 – 25 = 5

4

Healthy batteries = 100 – 30 = 70

FP = 10

TN = 70 – 10 = 60

Scenario 2: Solar Panel Defect Detection

An AI system classifies solar panels as **Defective** or **Good**.

Out of 200 panels:

- 40 panels are actually defective.
- The model correctly detects 35 defective panels.
- 15 good panels are incorrectly classified as defective.

Q1. Find TP.

Q2. Find FN.

Q3. Find FP.

Q4. Find TN.

Answer

	Predicted Defective	Predicted Good
Actual Defective	35	5
Actual Good	15	145

Answers:

1.

Answer: 35

2. $40 - 35 = 5$

3. **Answer: 15**

4. Good panels = $200 - 40 = 160$

TN = 160 – 15 = 145

Scenario 3: Water Quality Classification

An IoT + ML system predicts whether water is **Unsafe** or **Safe**.

The confusion matrix is:

	Predicted Unsafe	Predicted Safe
Actual Unsafe	80	20
Actual Safe	10	90

Q1. What is TP?

Q2. What is FN?

Q3. What is FP?

Q4. What is TN?

Q5. What is the accuracy?

Scenario 4: Disease Detection

An AI model predicts whether a patient has a particular disease.

	Predicted Disease	Predicted No Disease
Actual Disease	95	5
Actual No Disease	10	90

Q1. What does TP = 95 mean?

Q2. What does FN = 5 mean?

Q3. What does FP = 10 mean?

Q4. What does TN = 90 mean?

Scenario 5: EV Charging Station Fault Detection

An AI system predicts whether an EV charging station requires maintenance.

Out of 500 charging sessions:

	Predicted Maintenance	Predicted No Maintenance
Actual Maintenance Required	120	30
Actual No Maintenance	20	330

Questions

Q1. Calculate accuracy.

Answer: 90%

Q2. Calculate precision.

Answer: 85.71%

Q3. Calculate recall.

Q4. What does the recall of 80% mean?

Scenario 6: Environmental Pollution Detection

An AI model predicts whether an area has **high air pollution**.

	Predicted High Pollution	Predicted Low Pollution
Actual High Pollution	70	30
Actual Low Pollution	15	85

Questions

Q1. What is the total number of observations?

Q2. Calculate accuracy.

Q3. Calculate precision.

Q4. Calculate recall.

Scenario 7: Battery Fire Risk Prediction

An ML model predicts whether an EV battery is at **high fire risk**.

Out of 1,000 batteries:

- TP = 45
- FP = 25
- FN = 5
- TN = 925

Questions

Q1. What does FN = 5 represent?

Q2. What is the accuracy?

Q3. What is the recall?

Q4. Why can accuracy alone be misleading in this scenario?

Quick Student Practice Questions

Q1

An AI model detects faulty motors. A motor is actually faulty and the model predicts faulty.

Q2

A healthy motor is predicted as faulty.

Q3

A faulty motor is predicted as healthy.

Q4

A healthy motor is predicted as healthy.

Q5

If $TP = 80$, $TN = 100$, $FP = 10$, and $FN = 10$, what is accuracy?

Q6

If TP = 90 and FN = 10, what is recall?

Q7

If TP = 80 and FP = 20, what is precision?

Q8

Which error occurs when an actually positive case is predicted as negative?

Q9

Which error occurs when an actually negative case is predicted as positive?

Q10

Which metric measures how many actual positive cases were correctly detected?

Key Formula Sheet

Metric Formula

Accuracy $(TP + TN) / (TP + TN + FP + FN)$

Precision $TP / (TP + FP)$

Recall $TP / (TP + FN)$

F1-Score $2 \times (Precision \times Recall) / (Precision + Recall)$

Simple memory trick:

- **TP:** Correctly found positive
- **TN:** Correctly found negative
- **FP:** False alarm

- **FN:** Missed positive case